

Divya Shanmugam

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Academic positions and degrees

Columbia University , New York, NY National Library of Medicine Fellow, Department of Biomedical Informatics	2026 –
Cornell Tech , New York, NY Postdoctoral Researcher, Department of Computer Science	2024 – 2026
Massachusetts Institute of Technology , Cambridge, MA Ph.D., Electrical Engineering and Computer Science (05/2024) Thesis: <i>Advancing Equity and Reliability in Machine Learning</i> .	2018 – 2024
Massachusetts Institute of Technology , Cambridge, MA M.Eng., Electrical Engineering and Computer Science (05/2018) Thesis: <i>Representation Learning for Improved Distance and Risk Metrics</i> .	2017 – 2018
Massachusetts Institute of Technology , Cambridge, MA B.S., Electrical Engineering and Computer Science (05/2017)	2013 – 2017

Honors

STAT Wunderkind	2026
Rising Star in EECS (MIT)	2025
Best Paper Award, Conference on Health, Inference, and Learning (CHIL)	2025
Honorable Mention Best Findings Paper, Machine Learning for Health (ML4H)	2023
Jane Street Fellowship (Honorable Mention)	2022
Best Talk, GW6 Research Summit	2021
National Science Foundation Graduate Research Fellowship	2017

Publications

* denotes equal contribution.

Auditing Human Labels with Concept-Based Decompositions D. Shanmugam, M. Zhang, A. Field, E. Pierson. <i>Under review</i> .	2026
Machine learning reveals hidden diagnoses among underserved patients D. Shanmugam, B. Hardy, A. Wang, S. Divikaran, E. Pierson**, M. Barnett**. <i>Under revision at JAMA Network Open; plenary talk at Society for General Internal Medicine</i> .	2026
Improving Spontaneous Labor Prediction with Electronic Health Record Data M. Krishnamoorthy*, D. Shanmugam*, D. Tjandra, A. Peahl, C. Pancaro, E. Ziedan, A. Kowalski, J. Wiens. <i>Under revision at Nature Communications Health</i> .	2026

Strategic Feature Selection and Regularization	2026
J. Kaur, P. Patil, D. Shanmugam, E. Pierson, M.I. Jordan, N. Haghtalab, M. Jagadeesan, A. Alaa, S. Wang. <i>NeurIPS 2026; Stellar Abstract Award at STAI-X.</i>	
Survival Analysis with Limited Overlap and Censoring Distribution Shift	2026
M. Krishnamoorthy, D. Tjandra, D. Shanmugam, A. E. Kowalski, J. Wiens. <i>MLHC.</i>	
Streamlining Oversight in Medicare Advantage: Measuring Coding Intensity via Contract Rankings	2026
T. Chang, M. Qiu, Z. Hu, D. Zapzalka, D. Shanmugam, D. K. Shenfeld, R. B. Parikh, M. Makar, E. J. Emanuel, Z. Brown, A. E. Kowalski, J. Wiens. <i>Health Affairs Scholar.</i>	
The Trillion Dollar Algorithm: Lessons from Machine Learning for Medicare Advantage Risk Adjustment	2026
D. Shanmugam*, M. Johnson*, D. Meyers, J. Wiens, E. Pierson. <i>Health Affairs Forefront.</i>	
A Roadmap for Addressing the Use of Race and Ethnicity in Clinical Algorithms	2026
D. Shanmugam*, S. Balachandar*, A. Chouldechova, J. Diao, K. Ferryman, A. Manrai, S. Pfohl, N. Powe, D. Raji, E. Pierson. <i>Nature Health.</i>	
Identifying Mechanisms of Disparities within Cascades of Cardiovascular Care after an Emergency Department Visit	2026
A. Wang, D. Shanmugam, S. Divikaran, E. Pierson, M. Barnett. <i>JAMA Health Forum.</i>	
Evaluating multiple models using labeled and unlabeled data	2025
D. Shanmugam*, S. Sadhuka*, M. Raghavan, J. Gutttag, B. Berger**, E. Pierson**. <i>Neural Information Processing Systems (NeurIPS) 2025.</i>	
Learning Disease Progression Models That Capture Health Disparities	2025
E. Chiang, D. Shanmugam, A. Beecy, G. Sayer, N. Uriel, D. Estrin, N. Garg, E. Pierson. <i>Conference on Health, Inference, and Learning (CHIL) 2025; Best Paper Award.</i>	
Test-time augmentation improves efficiency in conformal prediction	2025
D. Shanmugam, H. Lu, S. Swaminarayan, J. Gutttag. <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025.</i>	
Generative Artificial Intelligence in Medicine	2025
D. Shanmugam, M. Agrawal, R. Movva, I. Y. Chen, M. Ghassemi, M. Jacobs, E. Pierson. <i>Annual Review of Biomedical Data Science, 2025.</i>	
Using large language models to promote health equity	2025
E. Pierson*, D. Shanmugam*, R. Movva*, J. Kleinberg*, et al. <i>New England Journal of Medicine AI, 2025.</i>	

Quantifying disparities in intimate partner violence: a machine learning method to correct for underreporting	2024
D. Shanmugam, K. Hou, E. Pierson. <i>npj Women's Health</i> , 2024.	
Longitudinal Changes in Desire and Attraction Among Women Who Have Discontinued Hormonal Contraceptives	2024
J. Gassen, S. Mengelkoch, D. Shanmugam, J. Pierson, A. van Lamsweerde, E. Benhar, S. E. Hill. <i>Hormones and Behavior</i> , 2024.	
Coarse race data conceals disparities in clinical risk score performance	2023
R. Movva*, D. Shanmugam*, K. Hou, P. Pathak, J. Gutttag, N. Garg, E. Pierson. <i>Machine Learning for Healthcare Conference (MLHC)</i> , 2023; <i>Honorable Mention for Best Findings Paper at ML4H Symposium</i> .	
The Relationship Between Photoperiod and Ovulation Rate: A Multi-Site Study Using NaturalCycles Data	2023
D. Shanmugam, M. Espinosa, J. Gassen, A. van Lamsweerde, J. Pearson, E. Behar, S. E. Hill. <i>Scientific Reports</i> , 2023.	
Kaleidoscope: Semantically-grounded, Context-specific Model Evaluation	2023
H. Suresh, D. Shanmugam, A. Bryan, T. Chen, A. D'Amour, J. Gutttag, A. Satyanarayan. <i>ACM CHI Conference on Human Factors in Computing Systems (CHI)</i> , 2023.	
Data Augmentation for Electrocardiograms	2022
A. Raghu, D. Shanmugam, E. Pomerantsev, J. Gutttag, C. Stultz. <i>ACM Conference on Health, Inference, and Learning (CHIL)</i> , 2022.	
Learning to Limit Data via Scaling Laws: A Computational Interpretation for the Legal Principle of Data Minimization	2022
D. Shanmugam, S. Shabanian, F. Diaz, M. Finck, A. Biega. <i>ACM Conference on Fairness, Accountability, and Transparency (FAccT)</i> , 2022.	
Better Aggregation in Test-Time Augmentation	2021
D. Shanmugam, D. Blalock, G. Balakrishnan, J. Gutttag. <i>IEEE/CVF International Conference on Computer Vision (ICCV)</i> , 2021; <i>Oral Presentation</i> .	
Multiple Instance Learning for ECG Risk Stratification	2019
D. Shanmugam, D. Blalock, J. Gutttag. <i>Machine Learning for Healthcare Conference (MLHC)</i> , 2019; <i>Oral Presentation</i> .	

Selected talks

Modeling Human and Algorithmic Behavior in Healthcare	
<i>NYU Computer Science Colloquium, New York, NY</i>	2026
<i>Yale Department of Biostatistics, New Haven, NY</i>	2026
<i>Columbia Department of Biomedical Informatics, New York, NYU</i>	2026

Reliable machine learning with health data

<i>Applied Data Science, Cornell Tech, New York, NY</i>	2025
<i>AI & Public Policy Seminar, Cornell Tech, New York, NY</i>	2025
<i>Wadhvani School of Data Science & AI Seminar, IIT Madras, Virtual</i>	2025

Methods to Evaluate the Quality of Clinical Care

<i>INFORMS, Atlanta, GA</i>	2025
<i>International Conference on Statistics and Data Science (ICSIDS), Vancouver, CA</i>	2025
<i>Center for Population Health (CPH), UC Berkeley, Berkeley, CA</i>	2025
<i>UVA School of Data Science, Charlottesville, VA</i>	2025
<i>Healthy ML Group, MIT, Cambridge, MA</i>	2025
<i>Diversity in Health Data: Achieving Benefit for All, Virtual</i>	2025

The Medicare Advantage Algorithm

<i>AI & Public Policy Seminar, Cornell University, Ithaca, NY</i>	2025
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Advancing Equity & Reliability in Machine Learning

<i>Progress Educational Trust, Virtual</i>	2025
<i>Statistical Methods for Health Equity, Data Science for Health Equity, Virtual</i>	2024

All the Data We Cannot See

<i>AI and Medicine Workshop, Mt. Sinai, New York, NY</i>	2024
<i>Computational Healthcare for Equity and iNclusion Lab, UC Berkeley, Berkeley, CA</i>	2023

Quantifying Inequality in Underreported Conditions

<i>MIT Algorithmic Fairness Reading Group, Cambridge, MA</i>	2023
<i>MIT AI Ethics Seminar, Cambridge, MA</i>	2022
<i>University of Chicago Crime and Education Lab, Virtual</i>	2021
<i>Cornell Information Science Seminar, Virtual</i>	2021
<i>Microsoft Research New England, Virtual</i>	2021

At the Intersection of Conceptual Art and Deep Learning: The End of Signature

<i>List Center, Cambridge, MA</i>	2022
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Learning to Limit Data Collection using Scaling Laws

<i>Microsoft Research Montreal, Virtual</i>	2020
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Machine Learning, Data Collection, and Women's Health

<i>Texas Christian University, Virtual</i>	2020
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Mentorship

James Kang (2026-Current); Ester Muzychuk (2026-Current); Erica Chiang (2024-2025); Helen Lu (2021-2024); William Hou (2022-2023); Anna Bryan (2021-2022); Tiffany Chen (2021-2022); Angela Zhang (2021-2022); Neha

Hulkund (2020–2021); Roshni Sahoo (2018–2020); Skylar Gordon (2018–2019); Xinyi Guo (2018–2019).

Professional Service

Leadership

Organizing Committee, Workshop Chair, MLHC	2025
Organizing Committee, Publicity Chair, FAccT	2025
Organizing Committee, Workflow Chair, ML4H	2023
Applied Machine Learning Seminar Organizer, MIT	2023
Undergraduate Mentorship Improvement Initiative, MIT	2020 – 2021
GW6 Event Coordinator, MIT	2018 – 2019
MIT AI Mentorship Program Coordinator, MIT	2018 – 2020

Reviewing

Neural Information Processing Systems (NeurIPS)	2020, 2026
Conference on Fairness, Accountability, and Transparency (FAccT)	2022 – 2026
Conference on Health, Inference, and Learning (CHIL)	2020, 2022
Machine Learning for Healthcare (MLHC)	2020 – 2023, 2026
Machine Learning for Health Symposium (ML4H)	2019 – 2022
International Conference on Machine Learning (ICML)	2021 – 2022
Computer Vision and Pattern Recognition (CVPR)	2021
Women in Machine Learning (WiML), NeurIPS Workshop	2018, 2020

Community Outreach

Reviews and Rebuttals Mentorship Panel (ML4H) — Panelist	2025
Round Table on Algorithmic Fairness (ML4H) — Discussant	2025
AI and Geriatrics Workshop (Johns Hopkins) — Invited participant	2025
AI and Society Panel (Olin College) — Panelist	2023
Career Mentorship Panel (MIT) — Panelist	2021
Graduate Student Panel (McCormick Hall) — Panelist	2020
Graduate Student Panel (MIT Women in EECS) — Panelist	2019
Lightning Talks (MIT Women in EECS) — Speaker	2017